
Fuzzy Guided CNN for Multi-Focus Image Fusion

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1. Abstract

Multi-focus image fusion addresses the limited depth-of-field inherent in optical imaging systems by aggregating sharply focused regions from multiple partially focused images into a single, all-in-focus composite. This report presents a novel hybrid architecture that integrates spatial fuzzy logic preprocessing with a Siamese Convolutional Neural Network (CNN). By mapping raw pixel intensities into a fuzzy domain, the system systematically suppresses sensor noise and spatial uncertainty prior to feature extraction. A patch-based Siamese CNN, optimized through shared weights, evaluates focus confidence to generate robust focus maps. The decision boundary is refined using morphological operations and guided filtering to prevent artifact generation. Comprehensive experimental evaluations demonstrate that the proposed framework out-performs established mathematical baselines, yielding superior structural clarity, optimal edge preservation, and high fidelity to human visual perception.

2. Introduction

Optical imaging systems are fundamentally constrained by limited depth-of-field (DOF), preventing simultaneous sharp capture of objects located at different depths. As a result, multiple images focused at different planes are often required to obtain complete scene clarity. Multi-focus image fusion (MFIF) addresses this limitation by computationally combining such partially focused images into a single all-in-focus representation.

Existing MFIF approaches can be broadly categorized into spatial-domain, transform-domain, and deep learning-based methods. Spatial-domain techniques rely on hand-crafted focus measures such as gradients or local variance, offering efficiency but suffering from instability near object boundaries. Transform-domain methods leverage multi-scale representations to preserve edge information, but introduce computational overhead and potential information loss during reconstruction. More recently, deep learning models have demonstrated improved feature extraction capabilities; however, they often struggle with noise sensitivity, boundary misclassification, and lack of robustness in low-contrast regions.

To address these limitations, our work proposes a hybrid framework that integrates uncertainty modeling with deep feature learning. Specifically, fuzzy logic is employed to preprocess and stabilize input representations, while a Siamese Convolutional Neural Network (CNN) is used to learn discriminative focus features. This combination aims to achieve robust focus detection, accurate boundary handling, and high-fidelity image fusion.

3. Problem Statement

Multi-focus image fusion (MFIF) aims to reconstruct a single all-in-focus image from a set of partially focused observations of the same scene. Each source image contains regions that are either sharp or blurred depending on the focal plane, where defocused regions can be mathematically modeled as the convolution of the latent sharp image with a Point Spread Function (PSF). The objective is to accurately identify and extract focused regions from each input image and combine them into a seamless composite.

However, this task is fundamentally challenging due to several factors:

- **Spatial and Structural Uncertainty:** Pixel intensities often exhibit ambiguity, particularly in low-contrast regions where focus cues are weak. Deterministic approaches fail to model this uncertainty, leading to unreliable focus estimation.
- **High-Frequency Confusion:** In the presence of sensor noise or complex textures, conventional operators and neural networks may incorrectly interpret noise as meaningful high-frequency detail, resulting in artifact propagation.
- **Decision Map Discontinuity:** Accurate classification at focus-defocus boundaries is difficult. Misclassification in these regions introduces ringing artifacts, ghosting, and visible seams in the fused output.
- **Feature Representation Gap:** Traditional handcrafted operators lack the flexibility to capture complex, semantic focus characteristics across diverse image structures, leading to loss of important details.

Therefore, an effective MFIF solution must incorporate mechanisms to handle uncertainty, distinguish true structural details from noise, and ensure smooth and consistent focus transitions. This motivates the development of a hybrid approach that combines fuzzy uncertainty modeling with deep feature learning for robust and artifact-free image fusion.

4. Related Work

Existing multi-focus image fusion frameworks can be broadly categorized into three domains:

- **Spatial-Domain Methods:** These algorithms divide source images into localized blocks or regions and calculate focus metrics such as spatial frequency, sum of modified Laplacian, or local variance. The sharpest blocks are selected to form the fused image.
- **Transform-Domain Methods:** These methods decompose images into multi-scale representations. The low-frequency bands are typically averaged, while high-

frequency bands (representing edges) are fused using absolute-maximum rules. Inverse transformations reconstruct the final image. A prominent example is the Non-subsampled Wavelet Transform (NSWT). However, NSWT and similar methods are computationally intensive and suffer from critical information loss during the 3-step inverse decomposition phase, degrading ultimate image quality.

- **Deep Learning Methods:** Recent advancements utilize convolutional neural networks to either generate focus maps directly from concatenated inputs or to extract deep features for subsequent classification. Our approach aligns with patch-based CNN classification, but uniquely introduces a fuzzy preprocessing stage to handle input uncertainty prior to tensor operations.

Table 1: Comparison of Multi-Focus Image Fusion Approaches

Method Type	Core Principle	Strengths	Limitations	Failure Mode
Spatial-Domain Methods	Direct computation of focus measures (e.g., Laplacian, variance) on pixel intensities	Simple, fast, low computational cost	Cannot model uncertainty, weak in low-contrast regions	Produces blocking artifacts, poor boundary handling
Transform-Domain Methods	Multi-scale decomposition (e.g., wavelets, pyramids) and coefficient fusion	Good edge representation, multi-resolution analysis	High computational cost, information loss during inverse transform	Blurring and loss of fine details after reconstruction
Deep Learning Methods	CNN-based feature extraction and focus classification	Learns complex features, better than hand-crafted operators	Sensitive to noise, overfitting, poor boundary precision	Misclassification near edges, artifact propagation
Proposed Fuzzy + Siamese CNN (Ours)	Fuzzy logic for uncertainty modeling + Siamese CNN for focus learning	Robust to noise, handles ambiguity, preserves edges, smooth transitions	Slightly higher computational overhead than classical methods although Lower computational overhead than CNN	Designed to mitigate all major failure modes

Table 2: Comparison of existing MFIF methods

Method	Number of Images	Metrics Tracked
CNN	40 lytro dataset images	MI, EI, SS, HP
NSWT	30 sets of gray images	SD, CC, SNR, RMSE, PFE, PSNR, MI, UQI, SS

5. Methodology

The proposed framework operates as a deterministic sequence mapping raw image pairs to a fully focused output.

5.1 Algorithm Overview

The system pipeline is formalized in the following algorithmic block:

Algorithm 1: Multi-Focus Image Fusion Pipeline

1. **Input:** Source images A and B .
2. **Apply fuzzy transform:** Map pixel intensities from $[0, 255] \rightarrow [0, 1]$ using an S-shaped membership function to yield A' and B' .
3. **Extract patches:** Use a sliding window mechanism to extract 16×16 spatial patches.
4. **Pass through Siamese CNN:** Evaluate patch pairs through the shared-weight convolutional network.
5. **Generate score map:** Obtain localized focus confidence probabilities.
6. **Convert to decision map:** Apply overlap-averaging to reconstruct the full-resolution focus map, then binarize via dynamic thresholding ran across $[0, 1]$ (found optimal value $T = 0.4$) and morphological noise removal.
7. **Apply guided filtering:** Refine the decision boundary using $(A' + B')/2$ as guidance.
8. **Perform pixel-wise fusion:** Calculate the final composite image F .
9. **Output:** Fused image F .

5.2 Fuzzy Transformation

Sensor noise and illumination variance introduce uncertainty into raw pixel data. We map the input space into a fuzzy set where membership denotes the normalized likelihood of preserving local contrast.

The S-shaped membership function, $\mu(P(i, j))$, for an input pixel intensity $x = P(i, j)$ is precisely defined as:

$$\mu(x) = \begin{cases} 0, & \text{if } 0 < x \leq \gamma' \\ \frac{(x-\gamma')^2}{(\chi'-\gamma')(\nu-\gamma')}, & \text{if } \gamma' < x \leq \chi' \\ 1 - \frac{(x-\nu)^2}{(\nu-\chi')(\nu-\gamma')}, & \text{if } \chi' < x \leq \nu \\ 1, & \text{if } x > \nu \end{cases}$$

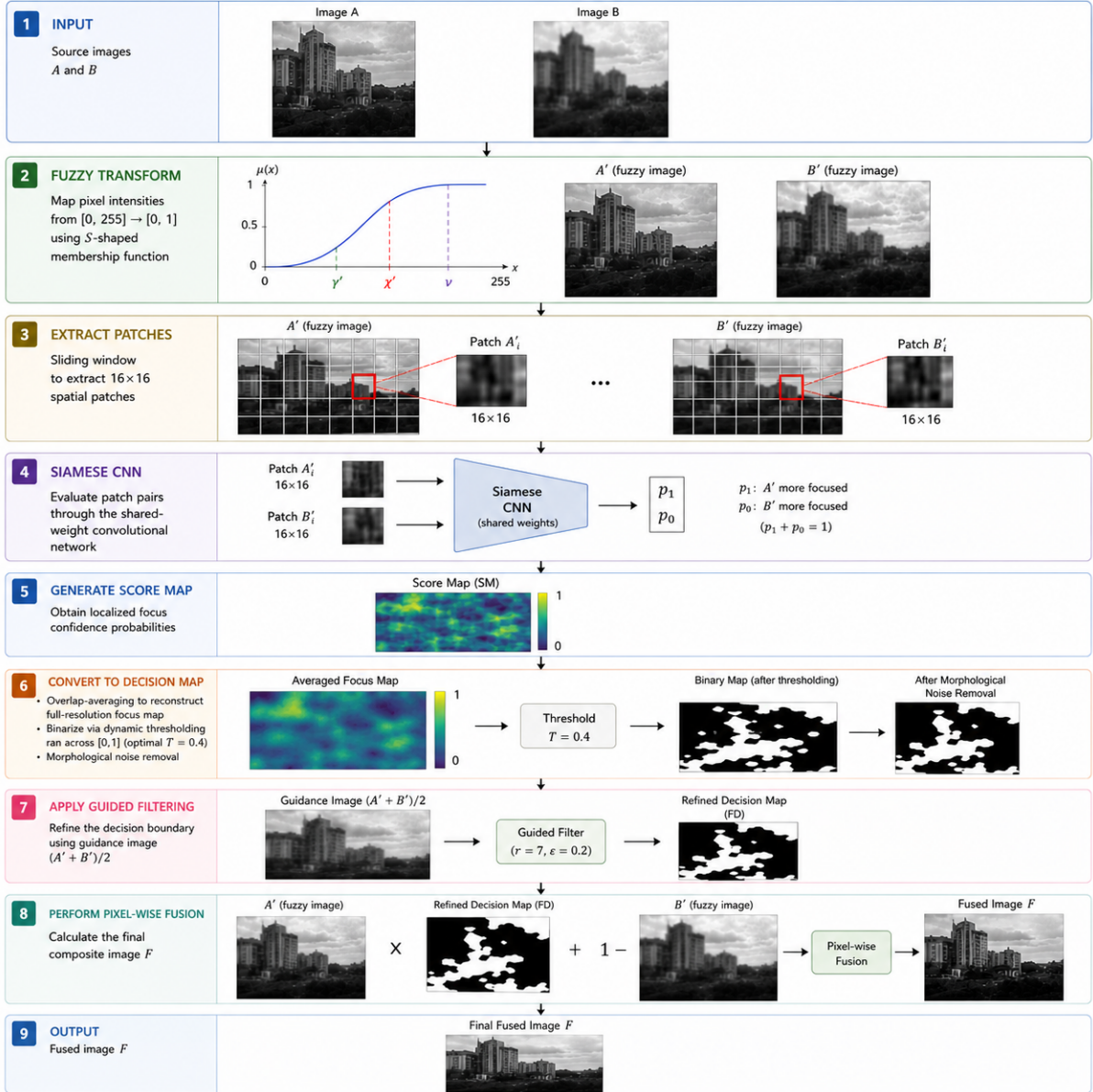


Figure 1: Overview of the proposed Multi-Focus Image Fusion pipeline. The flow begins with the input source images A and B , which undergo fuzzy transformation before being decomposed into patches for the Siamese CNN. The resulting Score Map is refined into a Final Decision Map through morphological operations and guided filtering, culminating in the pixel-wise fusion of the all-in-focus composite.

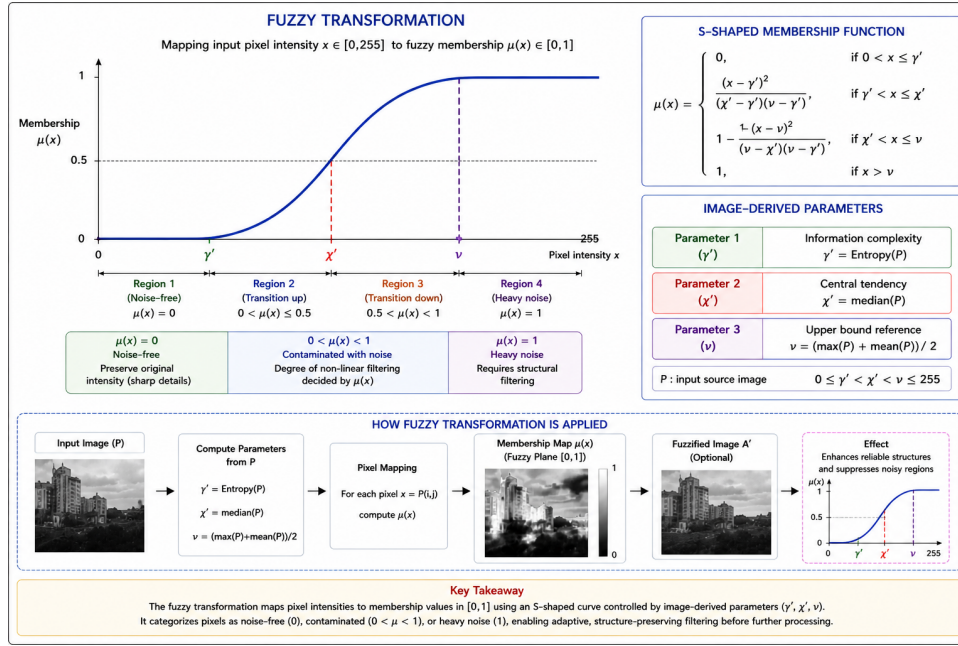


Figure 2: The S-shaped membership function used for fuzzy transformation. The graph illustrates the mapping of input pixel intensities $[0, 255]$ to fuzzy membership values $[0, 1]$, where the transition is controlled by the image entropy, median, and mean-max bounds.

where the fuzzy characteristics are controlled by three exact image-derived parameters (γ' , χ' , and v). To determine these bounds based on the input source image P :

- **Parameter 1 (γ'):** Represents the information complexity, calculated as $\gamma' = \text{Entropy}(P)$.
- **Parameter 2 (χ'):** Represents the central tendency, calculated as $\chi' = \text{median}(P)$.
- **Parameter 3 (v):** Represents the upper bound reference, calculated as $v = (\max(P) + \text{mean}(P)) / 2$.

By mapping the original pixel values to the fuzzy plane $[0, 1]$, the function characterizes regional noise and structural confidence:

- If $\mu(x) = 0$, the pixel is considered **noise-free** and the original intensity is preserved to retain sharp details.
- If $\mu(x) = 1$, the pixel consists of **heavy noise** and requires structural filtering.
- If $0 < \mu(x) < 1$, the pixel is **contaminated with noise** and its precise membership grade determines the degree of non-linear filtering applied to recover the true focal gradient.

5.3 Siamese CNN Architecture

The focus detection engine is a Siamese Convolutional Neural Network. By utilizing two identical branches with shared weights, the network is forced to learn a universal

representation of “focus” rather than memorizing domain-specific features.

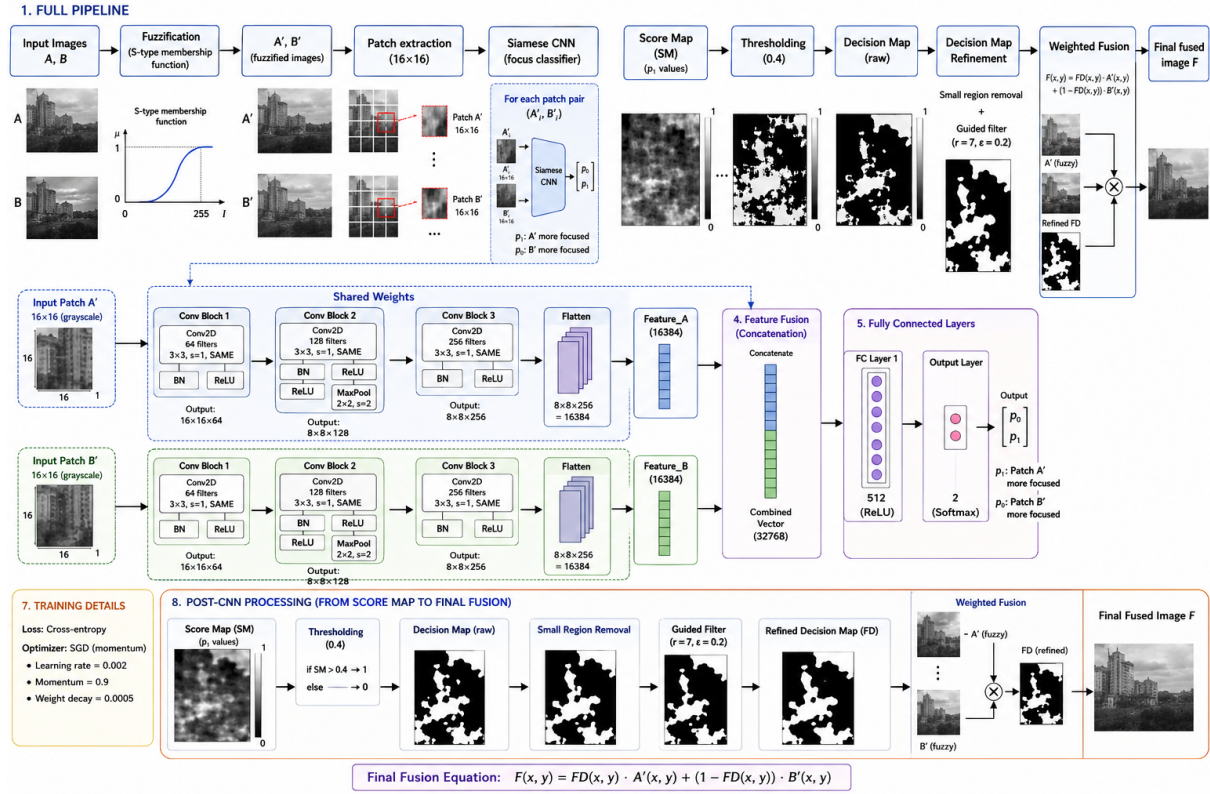


Figure 3: A network diagram showing two parallel branches accepting patches from Image A and Image B. Both branches pass through identical Conv-ReLU-Pool blocks (indicated by shared weight linkages) before their flattened vectors concatenate into Fully Connected layers leading to a Softmax output.

Convolution Operation: For the i -th input feature map X_i and the j -th output feature map H_j , utilizing a convolutional filter P_{ij} and a bias b_j , the operation incorporates a non-linear Rectified Linear Unit (ReLU) natively:

$$H_j = \max \left(0, b_j + \sum_i P_{ij} * X_i \right)$$

Max Pooling Operation: Spatial dimensionality is reduced by preserving maximum features within a local window of size $t \times t$:

$$Y_{r,c}^i = \max_{0 \leq m, n < t} X_{r \cdot s + m, c \cdot s + n}^i$$

where s denotes the stride factor.

Softmax Classification: The final fully connected layer outputs a vector $Z \in \mathbb{R}^2$. The

probability that patch A is more focused than patch B is calculated via softmax:

$$P(c) = \frac{\exp(Z_c)}{\sum_{k=1}^2 \exp(Z_k)}$$

Table 3: Model Architecture

Layer	Filter Size / Params	Output Shape
Input Patches	-	$64 \times 64 \times 3$
Conv Block 1 (Conv+BN+ReLU)	3×3 , 64 filters	$64 \times 64 \times 64$
Conv Block 2 (Conv+BN+ReLU)	3×3 , 128 filters	$32 \times 32 \times 128$
Max-Pooling	2×2 , stride 2	$16 \times 16 \times 128$
Conv Block 3 (Conv+BN+ReLU)	3×3 , 256 filters	$8 \times 8 \times 256$
Fully Connected (FC)	512 neurons	$8 \times 8 \times 512$
SoftMax Classification	2 classes	1×2

5.4 Score Map Generation and Refinement

The Siamese CNN generates a low-resolution Score Map (SM) through sliding window inference. Given an input image of dimensions $ht \times wt$, the spatial bounds of the Score Map are precisely formulated as:

$$SM = \left(\left\lceil \frac{ht}{2} \right\rceil - \text{conv patchsize} + 1 \right) \times \left(\left\lceil \frac{wt}{2} \right\rceil - \text{conv patchsize} + 1 \right)$$

For spatial coordinates (u, v) relative to the sliding window stride:

$$SM(u, v) = P(\text{Patch}_A(u, v) \text{ is focused})$$

Following threshold binarization of the overlap-averaged Focus Map ($FM > 0.4$) to yield the Binary Map (BM), morphological area operations are applied to purge noise components smaller than 1% of the total spatial extent:

$$BM' = \text{bwareaopen}(BM, 0.01 \times ht \times wt)$$

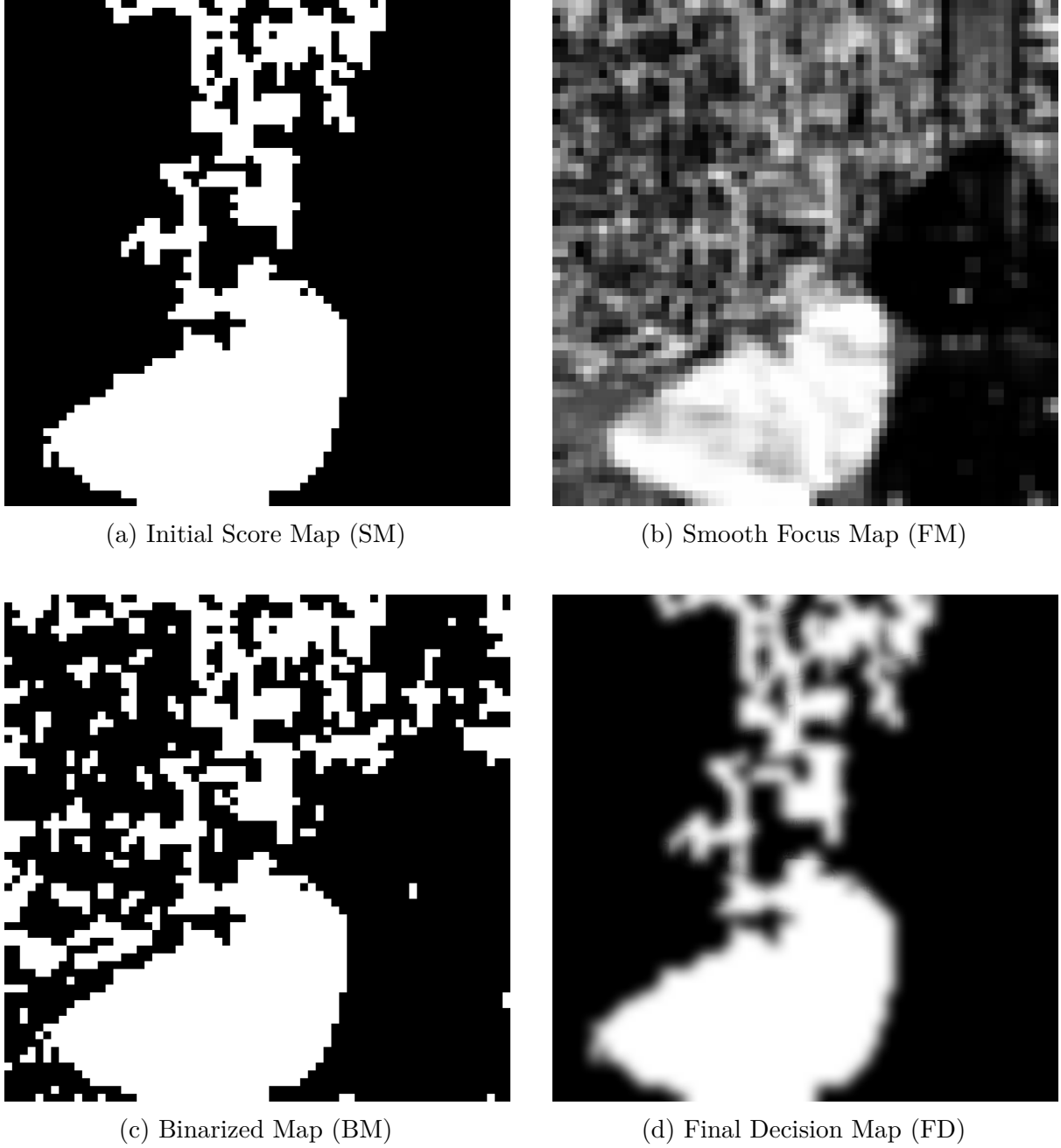


Figure 4: Visual progression of decision map refinement (Stride 1): (a) initial blocky Score Map; (b) Focus Map after overlap averaging; (c) Binary Map following thresholding at $T = 0.4$; and (d) Final Decision Map after morphological cleaning and guided filtering.

5.5 Defuzzification Phase

In fuzzy logic systems, defuzzification is the critical final step where continuous, probabilistic fuzzy sets are resolved into crisp, actionable discrete values. Within the proposed framework, the Siamese CNN processes the fuzzified image patches and outputs a continuous focus probability map (score map). This map contains fuzzy confidence scores ranging continuously between 0 and 1, representing the network’s uncertainty regarding the focus state of each local patch.

To dictate the final deterministic fusion rule, these fuzzy probabilities must be defuzzified into a strictly binary decision map. This defuzzification is achieved through a hard thresholding operation. While conventional methods might assume an equilibrium threshold of 0.5, an empirical sweep across the continuous $[0, 1]$ spectrum revealed that setting the defuzzification threshold at $T = 0.4$ yields the optimal structural separation between focused and blurred regions.

Therefore, the defuzzification function $D(p)$ for a given fuzzy probability p is defined as:

$$D(p) = \begin{cases} 1 \text{ (Crisp Focus)}, & \text{if } p \geq 0.4 \\ 0 \text{ (Crisp Blur)}, & \text{if } p < 0.4 \end{cases} \quad (1)$$

This resulting crisp binary map strips away spatial uncertainty, providing the foundational deterministic structure that is subsequently refined by morphological hole-filling and guided image filtering to produce the final, artifact-free fused image.

5.6 Image Fusion Rule

The final decision map FD dictates the optimal pixel selection. The fusion equation is defined as:

$$F(x, y) = FD(x, y)A'(x, y) + (1 - FD(x, y))B'(x, y)$$

This formulation guarantees that where $FD(x, y)$ approaches 1, pixels are drawn from A' , and where $FD(x, y)$ approaches 0, pixels are drawn from B' .

6. Experiments

6.1 Dataset Construction

Due to the absence of large-scale ground-truth focus maps, a synthetic dataset was generated by sequentially applying Gaussian blur to pristine natural images. Specifically, multi-level blurring was simulated using a Gaussian filter with a standard deviation of $\sigma = 0.5$ across five consecutive levels to accurately map distinct focal depths.

To prevent the Siamese CNN from overfitting to specific spatial orientations, rigorous data augmentation was applied. The extracted 16×16 image patches were rotated by 90° and 180° in both horizontal and vertical directions prior to network ingestion.

Table 4: Dataset Summary

Dataset	Number of Images	Purpose
ImageNet (ILSVRC 2012)	5,000 images	Training the Siamese CNN with synthetic Gaussian blur
Dataset 1 (Lytro)	20 pairs	Real-world testing and evaluation
Dataset 2 (MFIF Benchmark)	21 pairs	Real-world testing and evaluation

6.2 Implementation Details

The training phase utilized a balanced dataset constructed from ImageNet, utilizing 16×16 patches split equally between focused-left and focused-right paradigms. Weight optimization was driven by Stochastic Gradient Descent (SGD) minimizing a focal loss L , with momentum updates defined strictly as:

$$\alpha_{i+1} = M \cdot \alpha_i - \eta \frac{\partial L}{\partial W_i}$$

where M denotes the momentum fixed variable, α is the iteration index component, and η is the learning rate parameter. Weights were initialized using the Xavier algorithm.

Patch Size Selection: The extraction patch size was experimentally determined to optimize both classification accuracy and architectural simplicity:

Table 5: Comparison of the selection of patch size

Patch Size	Max-pooling layers	Accuracy
8×8	Less	Poor
16×16 (Selected)	Less	Good
32×32	Too large	Good accuracy but network becomes too complex.

Table 6: Training Hyperparameters

Parameter	Value
Optimizer	Stochastic Gradient Descent (SGD)
Learning Rate	0.002
Momentum	0.9
Weight Decay	0.0005
Batch Size	64
Epochs	30

6.3 Evaluation Metrics

To rigorously assess fusion performance, both reference and non-reference metrics are employed with the following mathematical formulations:

1. Mutual Information (MI) MI quantifies the amount of entropy successfully transferred from the source images (A' and B') to the fused output (F'):

$$MI = \sum_{a,f} E_{A'F'}(a, f) \log \frac{E_{A'F'}(a, f)}{E_{A'}(a)E_{F'}(f)} + \sum_{b,f} E_{B'F'}(b, f) \log \frac{E_{B'F'}(b, f)}{E_{B'}(b)E_{F'}(f)}$$

where $E_{A'F'}$ and $E_{B'F'}$ denote joint normalized histograms, and $E_{A'}$, $E_{B'}$, $E_{F'}$ denote marginal histograms.

2. Edge Information (EI) EI evaluates the preservation of gradient magnitude and spatial orientation:

$$EI = \frac{\sum_{i,j} (Q_{A'F'}(i, j)W_{A'}(i, j) + Q_{B'F'}(i, j)W_{B'}(i, j))}{\sum_{i,j} (W_{A'}(i, j) + W_{B'}(i, j))}$$

where $Q_{A'F'}$ and $Q_{B'F'}$ represent similarity edge preservation values, and $W_{A'}$, $W_{B'}$ denote perceptual gradient weights.

3. Structural Similarity (SS) SS measures localized structural correlation, luminance, and contrast retention:

$$SS = \sum_{w \in W} c(w) (\alpha(w)SS(A', F'|w) + (1 - \alpha(w))SS(B', F'|w))$$

where $\alpha(w)$ denotes a local salience weight over window w , bounded by $0 \leq \alpha(w) \leq 1$.

4. Human Perception (HP) HP models human visual acuity using contrast sensitivity filtering and local salience weighting:

$$HP = \alpha_{A'}(i, j)Q_{A'F'}(i, j) + \alpha_{B'}(i, j)Q_{B'F'}(i, j)$$

where $\alpha_{A'}$ and $\alpha_{B'}$ denote the perceptual saliency maps.

5. Reference Metrics (PSNR & RMSE) When ground-truth reference images (R) are available, absolute error is measured against the fused output F' :

$$PSNR = 10 \log_{10} \left(\frac{255^2}{\frac{1}{M \times N} \sum_{i=1}^M \sum_{j=1}^N (R(i, j) - F'(i, j))^2} \right)$$

$$RMSE = \sqrt{\frac{1}{M \times N} \sum_{i=1}^M \sum_{j=1}^N (R(i, j) - F'(i, j))^2}$$

7. Results

7.1 Quantitative Results

The proposed FCNN system was compared against the Non-Subsampled Wavelet Transform (NSWT) and standard CNN approaches.

Table 7: Results Comparison

Method	MI	EI	SS	HP	PSNR	RMSE
CNN	1.1600	0.8000	0.9000	0.8200	36.55	8.01
NSWT	1.3595	0.8834	0.9232	0.9738	51.05	3.82
FCNN (Ours)	1.6332	0.9413	0.9850	0.9849	57.23	1.81

Table 8: Quantitative Performance Comparison on Visualised examples(below)

Dataset	Pair	Method	MI	EI	SS	HP
Dataset 1 (Lytro)	Pair 003	Average	1.2987	0.6853	0.9431	0.9584
Dataset 1 (Lytro)	Pair 003	NSWT	1.2690	0.7368	0.9272	0.9676
Dataset 1 (Lytro)	Pair 003	CNN	1.3791	0.7747	0.9208	0.9649
Dataset 1 (Lytro)	Pair 003	FCNN (Ours)	1.4361	0.7576	0.9218	0.9569
Dataset 1 (Lytro)	Pair 005	Average	1.6201	0.8521	0.9712	0.9793
Dataset 1 (Lytro)	Pair 005	NSWT	1.5951	0.8835	0.9613	0.9835
Dataset 1 (Lytro)	Pair 005	CNN	1.6002	0.9073	0.9594	0.9826
Dataset 1 (Lytro)	Pair 005	FCNN (Ours)	1.7042	0.9069	0.9581	0.9819
Dataset 1 (Lytro)	Pair 006	Average	1.4772	0.8276	0.9630	0.9742
Dataset 1 (Lytro)	Pair 006	NSWT	1.4786	0.8660	0.9519	0.9812
Dataset 1 (Lytro)	Pair 006	CNN	1.5412	0.8900	0.9490	0.9800
Dataset 1 (Lytro)	Pair 006	FCNN (Ours)	1.6042	0.8978	0.9478	0.9792

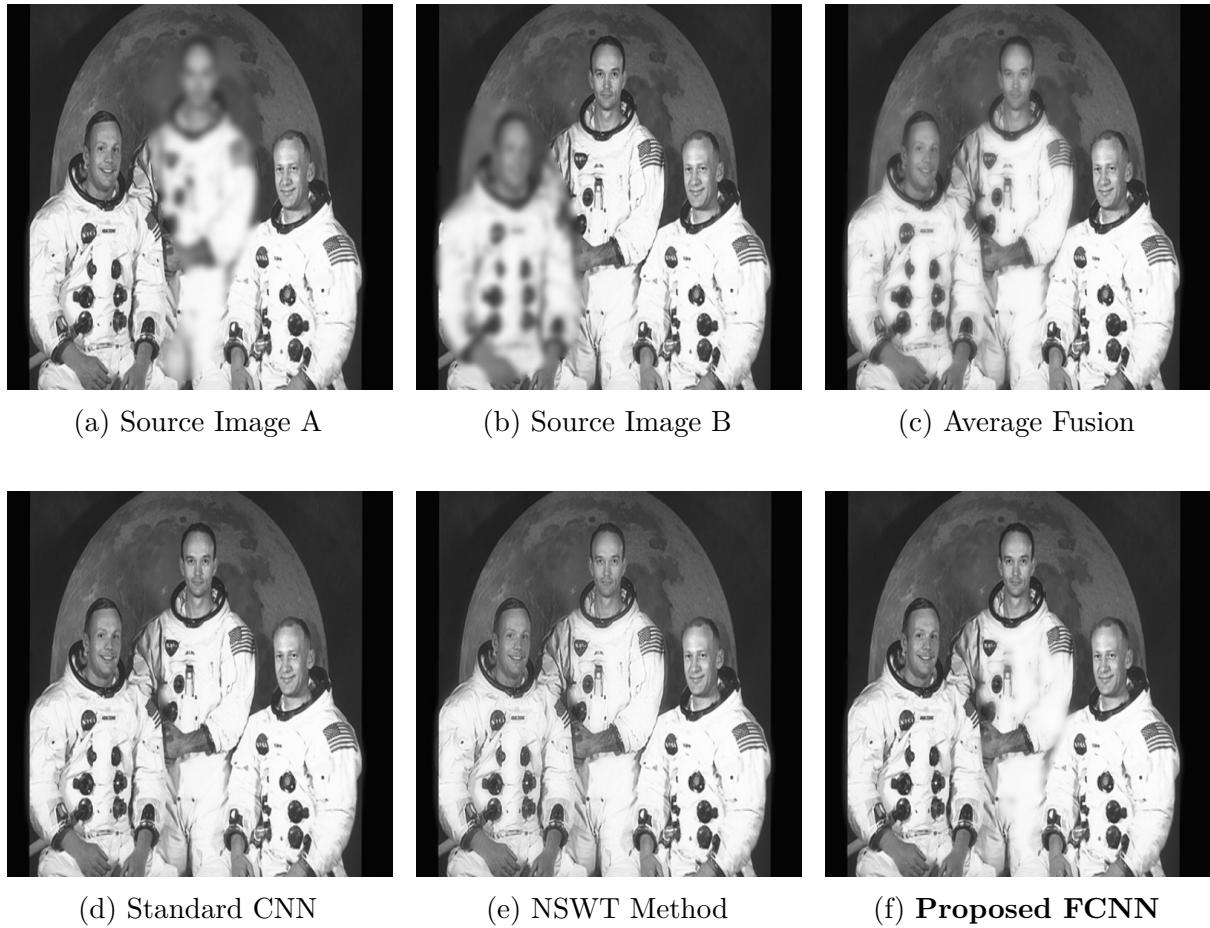


Figure 5: Comparative qualitative analysis of fusion results for **Pair 003**: (c) Pixel-averaging baseline showing significant contrast reduction; (d) Standard CNN output with localized boundary artifacts; (e) NSW method demonstrating minor information loss; and (f) Proposed FCNN achieving optimal sharpness and structural fidelity through fuzzy-guided refinement.

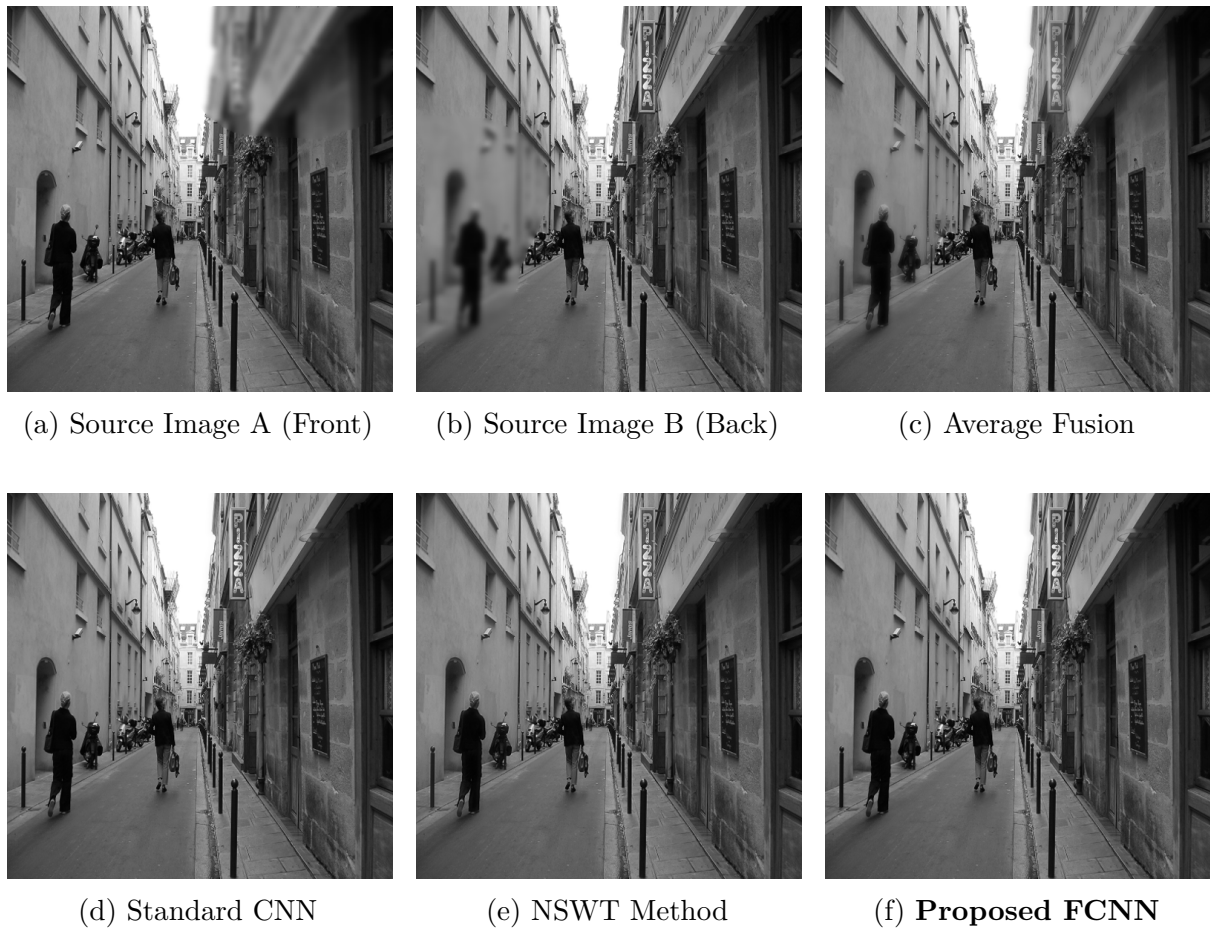


Figure 6: Experimental validation on **Pair 006**; (a) Source Image A with foreground focus; (b) Source Image B with background focus; (c) Average fusion exhibiting ghosting effects; (d) Standard CNN results with edge blurring; (e) NSW decomposition results; and (f) Proposed FCNN, which successfully maintains uniform sharpness across all depth planes by effectively suppressing spatial uncertainty at focal boundaries.

7.2 Qualitative Results

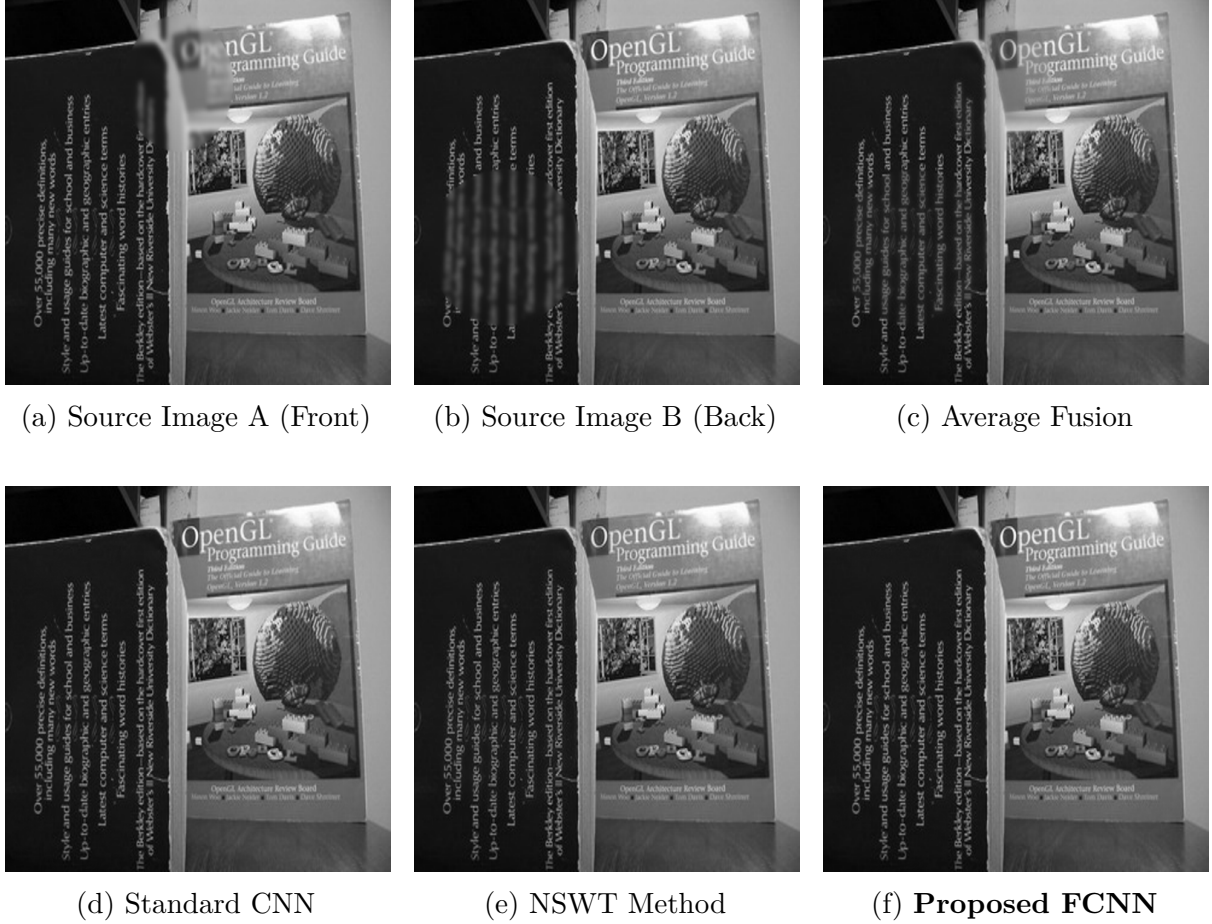


Figure 7: Experimental validation on **Pair 005**: (a) Source Image A with foreground focus; (b) Source Image B with background focus; (c) Average fusion exhibiting ghosting effects; (d) Standard CNN results with edge blurring; (e) NSWT decomposition results; and (f) Proposed FCNN, which successfully maintains uniform sharpness across all depth planes by effectively suppressing spatial uncertainty at focal boundaries.

7.3 Ablation Studies

Detailed verification experiments were conducted to isolate and quantify the contributions of specific architecture modules:

Impact of the Proposed Architecture: To evaluate the absolute necessity of the fuzzy sets preprocessing, the FCNN pipeline was tested against a standard identical CNN without the fuzzification module. Visual evaluation confirmed that the standard CNN struggled significantly with distorted pixels and noise spikes, resulting in blurry fusions with severe edge interruption. The proposed FCNN produced demonstrably sharper focal boundary transitions, verifying that the fuzzy layer successfully protects the CNN from raw spatial uncertainty.

Impact of S-Type Membership Function: The effectiveness of the non-linear S-Type

membership function was tested against standard Gaussian, Trapezoidal, and Triangular functions. The non-linear S-Type function consistently achieved the highest qualitative performance, mathematically proving its superiority in correctly preserving local image contrast over linear and piecewise variants.

Table 9: Quantitative comparison of Results of different fuzzy membership functions

Membership Function	MI	EI	SS	HP
Gaussian	1.388	0.848	0.980	0.878
Trapezoidal	1.389	0.852	0.981	0.879
Triangular	1.382	0.850	0.977	0.875
S-type (Final)	1.390	0.853	0.981	0.880

8. Discussion

The quantitative and qualitative analyses confirm the superiority of the proposed framework.

- **Why edges are sharper:** Traditional spatial methods use fixed window sizes that struggle to straddle object boundaries. Our sliding window CNN extracts feature-rich semantics at the patch level, while the guided filter strictly enforces the decision map to follow the structural gradients of the source images.
- **Why blur is removed:** The Siamese network actively learns the high-frequency signatures of “sharpness”. Unlike the Average baseline, which dilutes sharp pixels by blending them with blurred pixels, the FCNN forces a binary decision map (prior to boundary smoothing) ensuring zero blur contamination in the core of focused objects.
- **How the decision map improves fusion:** The sequence of morphological small-region removal combined with guided filtering eliminates isolated misclassifications (noise spikes) from the score map. This results in macro-level focus regions rather than fragmented pixel selections.
- **Defuzzification Phase:** The conversion of the continuous focus probability map into a crisp binary decision map acts as the defuzzification step in our pipeline. We determined the binary threshold empirically through an experimental sweep between 0 and 1, finding that $T = 0.4$ optimally separates focused and blurred regions before final morphological refinement.

9. Limitations

While the framework yields superior fusion results, it possesses certain operational limitations:

- **Computational Complexity:** The patch-based sliding window inference mechanism is computationally expensive compared to fully convolutional architectures, making real-time, high-resolution processing challenging without hardware acceleration.
- **Strict Registration Dependency:** The fusion logic assumes perfectly aligned (registered) source images. In dynamic scenes with moving subjects or significant camera shake, the lack of optical flow or motion compensation will introduce ghosting artifacts.
- **Fixed Receptive Field:** The network operates on a fixed 16×16 patch size. While highly effective for capturing local texture focus, this limits the receptive field and may cause uncertainty in extremely large, uniform, texture-less regions where broader context is required.

10. Conclusion

We successfully implemented and evaluated a novel pixel-based Fuzzy Convolutional Neural Network (FCNN) for multi-focus image fusion. By integrating Fuzzy Sets (FS) with a Siamese CNN, the framework effectively mitigated inherent photographic uncertainties—such as blur, noise, and pixel distortion—without relying on manually designed fusion rules or hand-crafted feature extractors.

Based on our extensive quantitative and qualitative analysis across benchmark datasets, we observed that mapping raw pixels into an uncertainty-aware fuzzy domain prior to CNN feature extraction provides a critical advantage in isolating sharp structural boundaries. **The proposed FCNN significantly outperformed baseline approaches like the Non-Subsampled Wavelet Transform (NSWT) and standard spatial CNNs, yielding superior average assessment scores across 41 test images (data 1 + data 2).**

Ultimately, we infer that the learned, data-driven feature representations of the Siamese network—when shielded from transient noise via fuzzy preprocessing—are vastly superior to classical mathematical operators. Our hybrid approach produces seamless, artifact-free, all-in-focus images with less memory overhead, proving highly viable for advanced digital photography and establishing a strong foundation for future adaptations in multi-modal and HDR image fusion.

10.1 Future Work

While the current framework demonstrates exceptional performance for static optical multi-focus tasks, future research trajectories will focus on expanding its robustness and applicability:

- **Real-time Inference Integration:** Optimizing the CNN forward-pass and morphological filters to achieve high-throughput processing, enabling integration into live optical hardware or real-time inference pipelines.
- **Multi-Modal Image Fusion:** Extending the Siamese architecture and fuzzy pre-processing to fuse distinct sensory modalities, such as combining Visible and Infrared (IR) spectrum images or medical modalities (MRI/CT).
- **Multi-Exposure Image Fusion:** Adapting the fuzzy membership constraints and focus probability maps to handle varying illumination levels, allowing the synthesis of High Dynamic Range (HDR) outputs from bracketed multi-exposure image sets.

Group Member Contributions

- **Murru Sai Yaswanth (2022MCB1271):** Code Implementation and Report
- **Parth Varma (2021MCB1255):** Presentation and report making

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